

Project Proposal: RealUltimate

Title: RealUltimate: Real-Time Statistics & Live Streaming for Ultimate Frisbee

Team Name: Team 6

Date: November 28th, 2025

Team Members: Jack McIntyre, Damien Rainbow Dever, Brehon Allen

1. Executive Summary

Ultimate Frisbee is one of the fastest-growing youth sports in the United States, yet the technology used to track stats and broadcast games remains archaic compared to established sports like baseball or basketball. Currently, coaches use older apps like iUltimate (UltiAnalytics) or paper and pencil to track complex game events, while parents and fans at home are often left in the dark, waiting for sporadic text updates to know the score. Apps like GameChanger exist for other sports and allow parents and fans to see score updates live and livestreaming, but they cannot accommodate the unique rules and flow of Ultimate Frisbee.

RealUltimate is our proposed mobile application designed to bridge this gap. Our system will allow teams to digitally manage rosters and record game statistics in real-time. The app will feature two distinct interfaces: a "Recorder" view for coaches to input data with single-tap buttons, and a "Spectator" view that allows fans to watch a live text-based play-by-play from anywhere in the world.

Our Minimum Viable Product (MVP) will focus on a high-performance, low-latency database that synchronizes the game score and possession data instantly. As a stretch goal, we intend to implement live video streaming, allowing parents to watch the game feed directly through the app. RealUltimate aims to synchronize the scorekeeping capabilities of iUltimate with the real-time spectator experience of GameChanger to professionalize the Ultimate Frisbee experience for players and fans worldwide.

2. Problem Statement

Context and Analysis

Ultimate Frisbee is a dynamic sport with unique statistical events (e.g., "callahans," "pulls," and rapid turnovers). The current landscape for tracking these events is fragmented and inefficient. According to the Sports & Fitness Industry Association, participation in Ultimate has consistently outpaced traditional field sports in recent years, yet the digital infrastructure has not kept pace.

1. **The Coach's Problem:** Existing apps like *iUltimate* utilize outdated user interfaces that are difficult to use on small screens in bright sunlight. However, the app does a good job handling the fast pace of points. There will be minor improvements in this area, but the big improvement will come on the fan side.
2. **The Fan's Problem:** There is no centralized "GameChanger" for Ultimate. Parents of high school and college players often cannot travel to every tournament. Without a live digital

feed, they are unable to follow the game's progress in real-time.

Customer Interview Summary

To validate this need, we conducted a customer interview with Jack McIntyre, who plays Ultimate Frisbee. He described the current limitations his team faces:

"When we travel for tournaments, my parents constantly text me asking for the score, and I can't answer because I am playing. We are too busy to send more than one score update at halftime via social media. We tried iUltimate, but there is no way for family and fans to view these in real time. We need something that is simple and instantly updates our fans at home."

Rationale

This feedback, combined with market data, confirms that the primary barrier is the lack of consistent, instant spectator updates. RealUltimate addresses this by prioritizing a simple "Recorder" interface that pushes data to a cloud database, solving the connectivity problem for families.

3. Design Objectives

The following objectives represent the high-level technical goals of the system.

1. **Intuitive Data Entry:** Design a user interface for "Recorders" that minimizes the number of taps required to record common game events, ensuring the user does not miss gameplay while looking at the screen.
2. **Real-Time Synchronization:** Implement a low-latency backend system that synchronizes game state (score, possession, time) between the Recorder and Spectators with a delay of less than 3 seconds.
3. **Role-Based Access Control:** Develop a secure authentication system that restricts access to team data. Spectators must have a specific "Access Code" to view statistics, real-time score updates, and live video feeds for a team.
4. **Live Video Integration (Stretch Goal):** Implement a video streaming capability that allows the Recorder device to broadcast video alongside the statistical data if network conditions allow.

4. Major System Requirements and Constraints

a) Major System Requirements

L1. User Management & Authentication

- **L1.1:** The system shall allow users to register an account using an email and password.
- **L1.2:** The system shall allow a user to create a Team and generate a unique 6-digit alphanumeric case-insensitive Access Code.
- **L1.3:** The system shall allow Spectator users to join a specific Team by entering a valid Access Code.

- **L1.4:** Spectators of a team shall be able to request the team owner for Recorder privileges.

L2. Recorder Operations

- **L2.1:** The interface shall prompt the Recorder to set the roster of active players, the number of timeouts allowed, the game-start, soft-cap and hard-cap times, and the points the game is to (13 or 15) before starting a Game.
- **L2.2:** The interface shall allow no more than 2 Recorders during one Game, one for stat-keeping, one for live-streaming (if time permits).
- **L2.3:** The interface shall prompt the Recorder to select 7 active roster players to play a point before starting the point.
- **L2.4:** The interface shall provide single-tap buttons for all offensive events: Goal, Drop, Throwaway, Callahan, Catch, and Injury (plus substitution).
 - **L2.4.1:** There shall be one Throwaway button and one Callahan button on the offensive interface.
 - **L2.4.2:** There shall be 7 Injury buttons on the offensive interface, one for each player.
 - **L2.4.3:** All buttons except Throwaway and Injury shall be duplicated for each player on the field except the player with the disc.
 - **L2.4.4:** Selecting Drop or Throwaway shall switch to the defensive interface, changing possession.
 - **L2.4.5:** Selecting Goal or Callahan shall end the point.
 - **L2.4.6:** Selecting Injury for a player shall prompt the recorder to select an active roster player not playing the current point to replace the injured player.
- **L2.5:** The interface shall provide single-tap buttons for all defensive events: Goal, Throwaway, D (defensive block), Callahan, and Injury (plus substitution).
 - **L2.5.1:** There shall be one Throwaway button, one Callahan button, and one Goal button on the defensive interface.
 - **L2.5.2:** There shall be 7 Injury and D buttons on the offensive interface, one for each player.
 - **L2.5.3:** Selecting D or Throwaway shall switch to the offensive interface, changing possession.
 - **L2.5.4:** Selecting Goal shall end the point.
 - **L2.5.5:** Selecting Injury for a player shall prompt the recorder to select an active roster player not playing the current point to replace the injured player.
 - **L2.5.6:** Selecting Callahan shall prompt the recorder to select the player that got the Callahan, and end the point
- **L2.6:** The system shall automatically update the game score after a point.
- **L2.7:** The interface shall provide a Timeout button that prompts the Recorder to select which team took the Timeout.
- **L2.8:** The interface shall provide an Undo button that reverts the last game or point state change (LIFO stack) to correct operator errors.

L3. Spectator Operations

- **L3.1:** The system shall display a Live Feed text log to all connected Spectators.

- **L3.1.1:** The Live Feed shall include all scoring events, turnover events, stoppage (timeout, halftime, game end) events, soft-cap, and hard-cap.
- **L3.2:** The Live Feed shall update automatically without requiring the user to refresh the screen.

L4. Live Video Streaming (Desired / Time Permitting)

- **L4.1:** The system shall allow the Recorder to enable "Live Stream Mode" utilizing the device's camera.

b) Major System Constraints

- **SC1 (Latency):** The synchronization delay between the input of an event and its appearance on a Spectator device shall not exceed 3 seconds on a standard 4G LTE connection.
- **SC2 (Platform):** If time permits, the application must be deployable on both iOS and Android mobile devices (Cross-platform). Otherwise, it must be deployed on Android.
- **SC3 (Visibility):** The data entry interface must use high-contrast color schemes (meeting WCAG AA standards) to remain visible in direct outdoor sunlight.
- **SC4 (Cost):** The operational cost for database hosting and authentication must remain within the "Free Tier" limits of the chosen cloud provider (Firebase) during the development phase.
- **SC5 (Hardware):** The system must function on standard consumer smartphones without requiring external cameras or sensors.

c) Standards

- **JSON (RFC 8259):** All data interchange between client and server shall follow standard JSON formatting.
- **TLS 1.3:** All network traffic shall be encrypted using Transport Layer Security to ensure data integrity.
- **OWASP Top 10:** The application authentication and data handling will strictly adhere to OWASP security standards to prevent injection attacks and broken access control.
- **WCAG 2.1:** User interface contrast ratios shall meet Web Content Accessibility Guidelines (Level AA) for outdoor visibility.

d) Societal Factors

- **Public Health:** By enabling remote spectatorship, the app reduces the pressure for parents to travel to tournaments while sick or during inclement weather, promoting public safety.
- **Social & Cultural:** The app helps bridge the gap for families of students studying out-of-state, allowing them to remain connected to the student's activities. It also democratizes access to "pro-level" stats for underfunded youth teams.
- **Environmental:** Moving from paper scorebooks to digital tracking reduces paper waste.
- **Economic:** As a free-to-use tool, this project removes the economic barrier for low-budget college clubs that cannot afford expensive subscription services like Hudl.
- **Global:** The app will help make ultimate a more global sport by expanding access for non-

US players to the same resources US players have.

5. Design Concept

a) Research and Investigation

We conducted a feasibility study comparing existing solutions:

- **GameChanger:** Excellent UI, but the game logic is hard coded for other sports and cannot be adapted for Ultimate.
- **iUltimate:** Was the best for this task. It was functional but saves data locally to the device. It did not offer real-time streaming to other phones, which is our primary differentiator. This app is also no longer supported and is not running.

Component List (Software Stack):

- **Frontend:** React Native (JavaScript/TypeScript)
- **Backend Database:** Google Firebase Realtime Database (NoSQL)
- **Auth Service:** Firebase Authentication SDK
- **Video Protocol:** WebRTC or HLS (HTTP Live Streaming)

b) Selection of Design Concept

We have selected a **Model-View-Controller (MVC)** architecture utilizing a Serverless Backend.

- **Frontend:** React Native. This ensures we satisfy Constraint C2 (iOS/Android support).
- **Backend:** Google Firebase Realtime Database. This NoSQL cloud database uses WebSockets to push updates to clients instantly, satisfying Objective 2 (Low Latency).

Figure 1: RealUltimate High-Level System Architecture

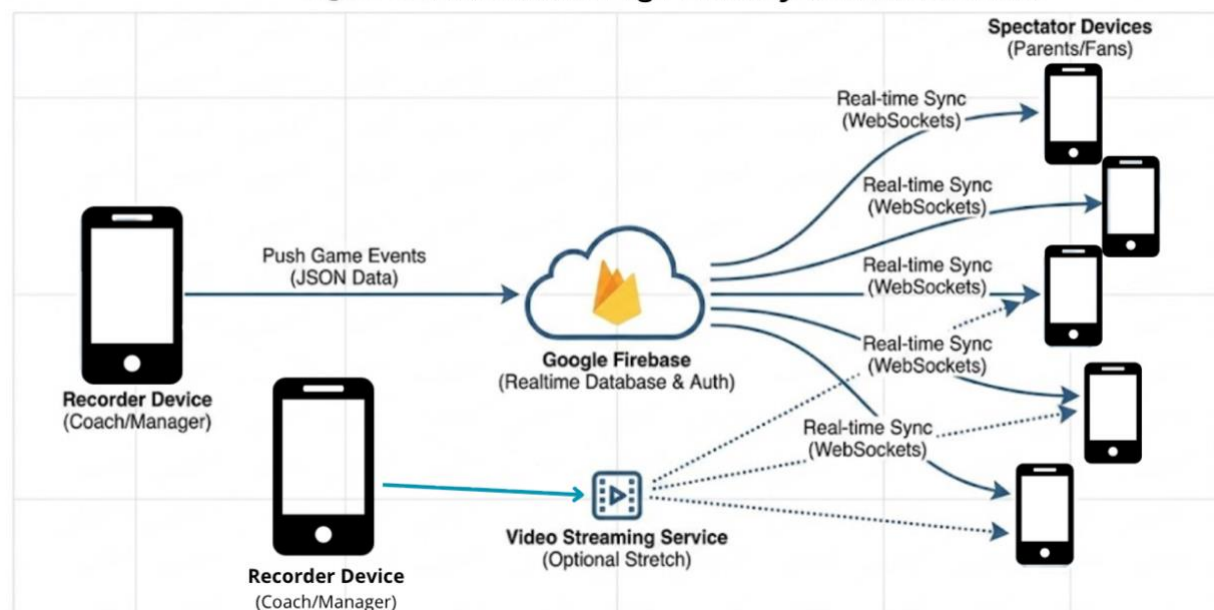


Figure 2: Initial Concept for “Recorder” Interface

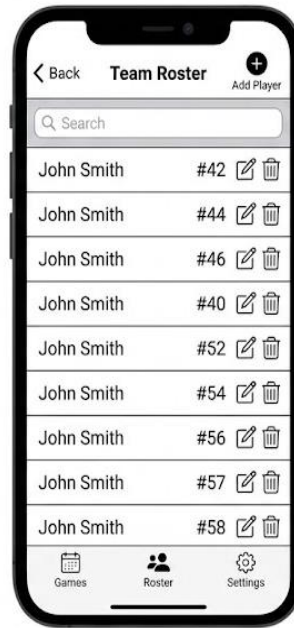


Figure 4: Concept for “Roster Management” Interface

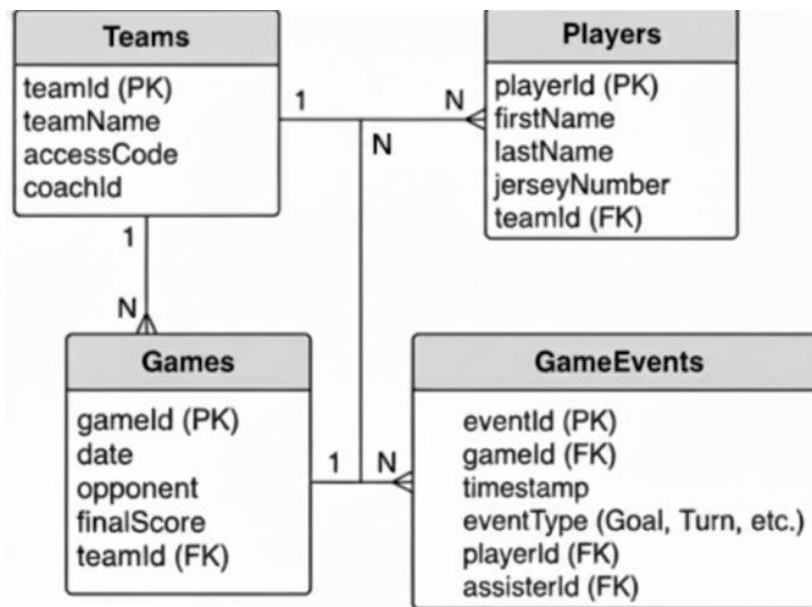


Figure 5: Firebase Database Schema

6. Deliverables and Milestones

a) Project Deliverables

The following items will be delivered to the customer and instructors:

1. **RealUltimate Mobile Application:** A functional installation file (.APK for Android / TestFlight for iOS).

2. **User Manuals:** Two PDF guides: "Coach's Guide" and "Spectator's Guide".
3. **Source Code:** Complete repository hosted on GitHub.
4. **Demo Video:** A video showcasing the app tracking a live scrimmage with side-by-side comparison of the real-time sync.

b) Schedule and Timeline

We will use an **Agile** framework with 2-week sprints. The Gantt chart below details our task breakdown.

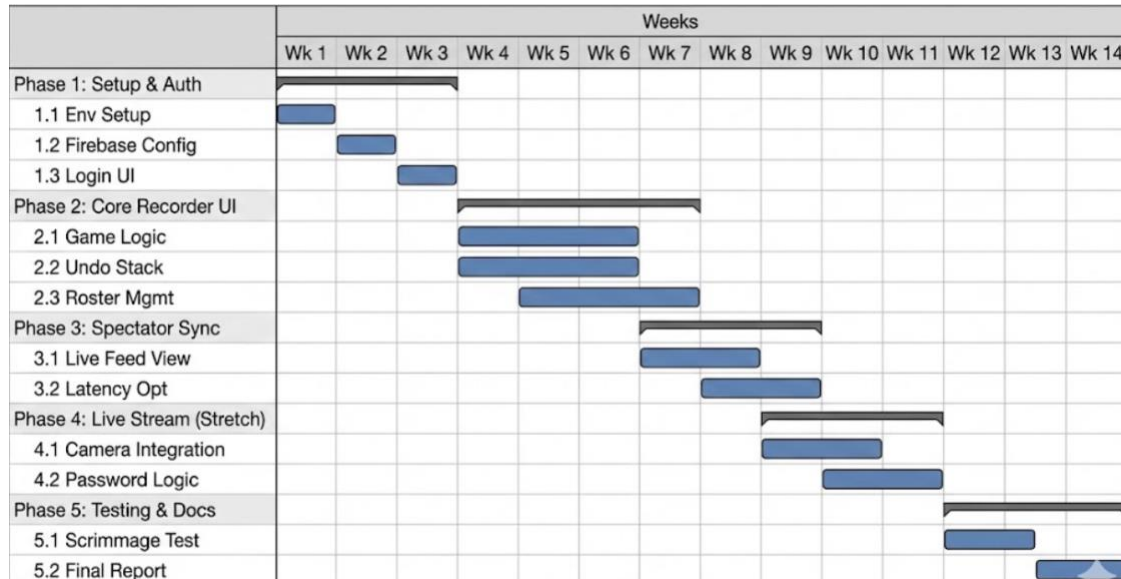


Figure 6: Project Gantt Chart (Overview).

Detailed Task List (Sub-levels):

- **Phase 1 (Weeks 1-3): Setup & Auth**
 - Task 1.1: Initialize React Native Environment [Jack McIntyre]
 - Task 1.2: Configure Firebase Project & API Keys [Brehon Allen]
 - Task 1.3: Build Login/Register Screens [Damien Rainbow Dever]
- **Phase 2 (Weeks 4-6): Core Recorder UI**
 - Task 2.1: Implement Game State Logic (Score/Possession) [Jack McIntyre]
 - Task 2.2: Implement "Undo" Stack Data Structure [Jack McIntyre]
 - Task 2.3: Build Roster Management Screen [Damien Rainbow Dever]
- **Phase 3 (Weeks 7-9): Spectator Sync**
 - Task 3.1: Create "Live Feed" List View [Damien Rainbow Dever]
 - Task 3.2: Optimize Database Listeners for Latency [Damien Rainbow Dever]
- **Phase 4 (Weeks 10-12): Live Stream (Stretch)**
 - Task 4.1: Camera Hardware Integration [Brehon Allen]
 - Task 4.2: Implement Stream Password Logic [Jack McIntyre]
- **Phase 5 (Weeks 13-14): Testing & Docs**
 - Task 5.1: Live Scrimmage Field Test [All]
 - Task 5.2: Final Report & Presentation [All]

7. Risk Management

a) Primary Risks to Project Success

1. **Video Latency:** Streaming video over cellular networks often incurs a 10-15 second delay. This could cause the "Live Text" to spoil the video (e.g., seeing "Goal" in text before it happens on video).
2. **Complex Game Logic:** Edge cases in Ultimate (e.g., a "Callahan" or a foul retracting a turnover) are difficult to model. A bug here could corrupt the score.
3. **UI Clutter:** Fitting a roster of 20+ players on a phone screen without making buttons too small is a major design challenge.

b) Risk Management Plan

- **Mitigation for Latency:** We will implement a programmed "delay" in the text feed if the user is watching video, ensuring the text and video stay in sync.
- **Self-Governing Strategy (Under-Contributors):** We will utilize a "Sprint Retrospective" every two weeks. If a member misses a deadline, they must pair-program with another member to catch up. If non-contribution persists, we can escalate to the course instructor. The other members can pick up any slack.

c) Contingency Plan

In the event of a campus closure (e.g., pandemic restrictions), our project is fully digital and can be completed remotely. We will simulate game data using a script rather than attending physical practices if necessary. All meetings will shift to Zoom/Discord.

8. Budget

This project is self-funded.

Item	Description	Estimated Cost
Database	Google Firebase (Spark Plan)	\$0.00
IDE	VS Code / Xcode	\$0.00
App Store Fee	Google Play Developer / Apple Developer Registration (One-time)	\$25.00 / \$99.00

Test Devices	Personal team devices	\$0.00
Web Domain	(Optional) realultimate.app	\$12.00
Total		\$25.00 - \$136.00

9. Team Considerations

a) Knowledge and Skills

- **Jack McIntyre (Computer Science and Engineering):** Experience with React Native. Will lead Frontend UI.
- **Damien Rainbow Dever (Computer Science and Engineering):** Strong SQL/NoSQL background. Will manage the Firebase Database.
- **Brehon Allen (Electrical Engineering):** Background in signals and networking. Will handle the Video Streaming protocols and latency testing.

b) Team Organization

We will meet weekly on Tuesdays at 5:00 PM. We use Trello for task management (Kanban style).

c) Interaction with Sponsor (Self-Sponsored)

As a self-sponsored project, we will all act as the "Product Owner." We are all responsible for defining the user requirements, Jack will certainly help the most with this based on his experience as a club player.

10. References

1. USA Ultimate, "Official Rules of Ultimate: 2024-2025 Edition," 2024.
2. Google Developers, "Firebase Realtime Database Documentation," 2024.
3. OWASP Foundation, "OWASP Top 10: 2021," 2021.
4. W3C, "Web Content Accessibility Guidelines (WCAG) 2.1," 2018.